

軟體測試

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授課時間：每週四 14：30—17：20

博理館 114

課程代號：EE 5170

識別碼：921 U9550

民國 107 年秋季班

期末考

考試中，請勿參閱資料。

In **Figure 1** is a procedure in C.

```
evenSum(a, b)
  int a, b;
{
  int sum = 0; k = b; count = 0;
  repeat
    if (k > sum-count && !(k % 3 || k % 5 ))
      sum=sum+k;
      k--;
    else
      k = k-2;
      count++;
  until k < a;
  if (sum > 3k)
    return (sum);
  else
    return 0;
} // evenSum( )
```

Figure 1: a simple C procedure

1. Draw the control-flow graph of the above program. (6pts/6)

- This is a closed-book exam.

4. Continuing from question 2, list the test requirements for prime-path coverage (PPC). (4pts/19)

5. Continuing from question 2, please write down a test set for the full PPC. (4pts/23)

6. Continuing from question 2, please list all of the du-paths with respect to k , *count*, and *sum*. (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/28)

7. Continuing from question 6, please write down a minimal test set that satisfies all-defs coverage with respect to k , *count*, and *sum*. (Direct tours only.) (5pts/33)

8. Continuing from question 6, please write down a minimal test set that satisfies all-uses coverage with respect to k , *count*, and *sum*. (Direct tours only.) (4pts/37)

9. The predicate of the if-statement in **Figure 1** is:

$$(k > \text{sum-count} \ \&\& \ ! (k \% 3 \ || \ k \% 5))$$

Please identify the clauses (*atoms* in standard logic terms) that determine this predicate.
(4pts/41)

10. Continuing from question 9, compute (and simplify) the conditions under which each of the clauses determines the predicate. (4pts/45)

11. Continuing from question 9, write the complete truth table for all clauses. Label your rows starting from 0 at the top. Use the format that row 0 should be all clauses true. You should include columns for the conditions under which each clause determines the predicate, and also a column for the predicate itself. (4pts/49)

12. Continuing from question 9, identify all pairs of rows from your table that satisfy *general active clause coverage (GACC)* with respect to each clause. (4pts/53)

13. Continuing from question 9, identify all pairs of rows from your table that satisfy *correlated active clause coverage (CACCC)* with respect to each clause. (4pts/57)
14. Continuing from question 9, identify all rows from your table that satisfy *implicant coverage (IC)* with respect to each implicant. (4pts/61)

15. Assume that we have a web program that collects shopping data with the following screen layout.

The GUI is a registration and query password form. It features a blue background with rounded corners. The form contains the following elements:

- Data item:** A text input field.
- Credit card #:** A text input field.
- Expiration year:** A text input field containing the value "2019".
- month:** A text input field containing the value "01".
- CVC:** A text input field.
- submit:** An oval button with the text "submit".
- refresh:** An oval button with the text "refresh".
- Close button:** A circular button with an "X" icon in the top right corner.

Figure 2: a GUI for registration and query password.

There is no other document. Please answer the following questions to properly test the page. Please define characteristics of inputs to the web interface and list them.
(4pts/65)

16. Continuing from question 15, please partition the characteristics into blocks and write them down. (4pts/69)

17. Continuing from question 16, please define and list values for the blocks. (4pts/73)

18. Assume that we have the procedure in **Figure 1** to test. Please write down a mutant with the Relational Operator Replacement (ROR) and write down a test case (values for a and b) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (4pts/77)

19. Assume that we have the procedure in **Figure 1** to test. Please write down a mutant with the Unary Operator Insertion (UOI) and write down a test case (values for a and b) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (4pts/81)

20. Continuing from Question 3, please write down the test cases (as pairs of parameter values) and prioritize them for the best efficiency of edge-pair coverage. (4pts/85)

21. What is Change-Impact-Analysis (CIA) in regression testing ? (3points/88)

22. Continuing from question 21, how can CIA be carried out ? (3points/91)

23. Please list some regression tools that you think you know and explain their functions.
(3points/94)

24. Please explain the oracle problem in software testing and its difficulties. (3points/97)

25. Continuing from question 24, please list some approaches toward the automation of test oracle. (3points/100)